

**The Geometry of Competent Contact:
What a Scalar Return Cannot See in an Agent's Grip on the
World**

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Abstract

Reinforcement learning scores a policy by a single number, the expected return $J(\pi) = \mathbb{E}[R]$, and treats everything an agent does as a means to that scalar. In competent action, such as a hand turning a cup to read its shape or a robot tapping a surface to tell metal from plastic, what improves is the agent’s grip on the world: the way its uncertainty about hidden states collapses into task-relevant structure. The agent’s belief over world-states is a sufficient statistic for control and lies on the probability simplex, where Chentsov’s theorem singles out the Fisher-Rao metric, so the movement of belief has a determined geometry. Competent contact is measured on that geometry by the contraction of uncertainty, the Fisher-Rao length of the belief trajectory, and the separation the terminal belief keeps between world-states. If reward depends on the world only through a partition g , the expected return depends on the belief only through its pushforward g_*b , so two beliefs at the maximal Fisher-Rao distance π can be reward-identical. In an exactly solvable active-perception POMDP, a reward-only policy and a competent-contact policy attain the same single-task return of 0.8 but leave terminal entropies of 1.09 and 0.68 nats; world-states differing only in the reward-irrelevant attribute collapse to Fisher-Rao distance 0.0 under the reward-only policy, while the competent policy keeps all six at least 2.44 apart. On a second task the first reward never mentioned, the reward-only agent scores at chance, 0.5, and the competent agent 0.85, a gap of 0.35 invisible to the single-task return.

Introduction

A reinforcement learner is defined by what it maximizes. The standard formulation takes the target to be a scalar, the expected sum of rewards $J(\pi) = \mathbb{E}[R]$, and rates the agent by the size of that number (Sutton and Barto 2018). Everything else, perception included, is instrumental. Sensing has value when it raises expected reward and none otherwise, and the quality of an agent’s contact with its world enters the theory only through its eventual effect on return.

The tradition that treated perception as an achievement described it differently. Gibson held that an animal actively picks up invariant structure in the ambient array, instead of receiving a retinal image and inferring a world from it, and that what it picks up are affordances, the possibilities the environment offers to that body (Gibson 1979). Merleau-Ponty described perception as tending toward an optimal hold on its object, the body settling into the world until the scene comes into the clearest grip it admits, and Dreyfus called that tendency a “maximal grip”, the drive to reduce the felt disequilibrium between perceiver and perceptual field (Merleau-Ponty [1945] 2012; Dreyfus 2002). O’Regan and Noë made the point operational: to see is to exercise mastery of sensorimotor contingencies, the lawful ways sensory input changes as the agent moves, so that vision is a skill of probing (O’Regan and Noë 2001). In these accounts perception is a competence, a graded quality of coupling between agent and world, measured by how well the coupling resolves the world into the structure the agent’s activity requires.

We formalize that idea. An agent’s competence is taken to be the geometry of how its actions transform its belief over the world, and this geometry is strictly richer than any scalar return com-

puted from it. Competent contact is a property of a trajectory on a curved manifold, and the return is a coarse projection of that trajectory onto a line. The sections below identify the structure the projection discards, prove that the loss is generic, and show on an exact model that the discarded structure is what allows an agent to carry competence from one task to another.

Belief space and the Fisher-Rao metric

The appropriate state variable for an agent that cannot observe the world directly is its belief about the world. Åström proved for a partially observed Markov process that the posterior distribution over hidden states is a sufficient statistic for optimal control, so an optimal policy can be written as a function of the belief alone (Åström 1965). Kaelbling, Littman, and Cassandra built the modern theory of planning on this fact, recasting a partially observable problem as a fully observable Markov decision process whose states are beliefs, so that the agent navigates the space of distributions over world-states (Kaelbling et al. 1998). That space is the probability simplex. A belief over N world-states is a point b in Δ^{N-1} , and an action, through the observation it yields and Bayes' rule, maps one point of the simplex to another.

The simplex carries a canonical metric. Rao observed that the Fisher information of a family of distributions defines a Riemannian metric and hence a geodesic distance between distributions (Rao 1945), and Chentsov proved its uniqueness: on a finite sample space the Fisher-Rao metric is, up to an overall scale, the only Riemannian metric invariant under the maps that carry one statistical model into another by a sufficient statistic (Chentsov [1972] 1982; the extension beyond finite sample spaces is Ay et al. 2017). Any other measure of distance between beliefs would introduce structure the statistics does not contain, so belief space is equipped with the Fisher-Rao geometry.

This geometry is explicit. The map $b \mapsto 2\sqrt{b}$ carries Δ^{N-1} isometrically onto part of a sphere of radius 2, on which the Fisher-Rao distance between two beliefs is the great-circle distance

$$d_{FR}(b, b') = 2 \arccos\left(\sum_i \sqrt{b_i b'_i}\right),$$

ranging from 0 for identical beliefs to π for beliefs with disjoint supports (Amari and Nagaoka 2000). A sphere of radius 2 has constant Gaussian curvature $1/4$. As a check on the numerical machinery, a curvature routine built only from the metric components and the Brioschi formula, with central finite differences, is applied to the Fisher metric at 36 grid points across a three-state marginal simplex; it returns 0.25 with a maximum deviation of 6.0×10^{-6} , the finite-difference error. With the space fixed, competence becomes a question about paths: how an agent's actions move its belief across this curved surface, how far the belief travels, and where it comes to rest.

Reward as a quotient of belief

The scalar return registers only part of the belief. A task assigns a reward $R(d, w)$ to each decision d the agent might commit to and each world-state w , and the Bayes value of a belief b is the best

expected reward it supports, $V(b) = \max_d \sum_w b(w)R(d, w)$. This value depends on the belief only through a quotient.

Proposition 1. *Suppose the reward depends on the world-state w only through a partition $g: W \rightarrow \{1, \dots, T\}$, so that $R(d, w) = \tilde{R}(d, g(w))$ for every decision d . Then the Bayes value $V(b) = \max_d \sum_w b(w)R(d, w)$ depends on the belief b only through its pushforward g_*b , the induced belief on classes. Consequently any two beliefs b, b' with $g_*b = g_*b'$ are reward-equivalent for the task, even when their Fisher-Rao distance $d_{FR}(b, b')$ is as large as the diameter π of the simplex.*

Proof. Write $c = g(w)$ for the class of w and group the sum by class: $\sum_w b(w)R(d, w) = \sum_w b(w)\tilde{R}(d, g(w)) = \sum_c (\sum_{w: g(w)=c} b(w))\tilde{R}(d, c) = \sum_c (g_*b)(c)\tilde{R}(d, c)$. The inner expression depends on b only through the class-sums $(g_*b)(c)$, so the maximum over d does too. If $g_*b = g_*b'$ then $V(b) = V(b')$. Beliefs supported on distinct states within a single class have disjoint support yet identical pushforward, and disjoint support gives $d_{FR} = \pi$. $\square$

The proposition is elementary. It states that the reward-relevant part of a belief is its projection onto the coarsest partition the reward respects, and that all finer structure is invisible to the return. Blackwell’s comparison of experiments expresses the same fact in terms of information: one experiment is sufficient for another when it is at least as useful in every decision problem, and an experiment can be sufficient for a particular task while discarding distinctions another task would need (Blackwell 1953). A scalar objective extracts from the belief its answer to one question, whereas the belief itself holds answers to all of them.

The simulation exhibits the extreme case. The world state is a pair $w = (A, B)$ of two independent attributes, $A \in \{0, 1, 2\}$ and $B \in \{0, 1\}$, giving 6 world-states on the simplex Δ^5 . Task 1 rewards depend on the world only through A . The two beliefs concentrated on $(A, B) = (0, 0)$ and on $(0, 1)$ have the same A -marginal, a point mass at $A = 0$, so both have task-1 value 1.0 and support the same optimal decision. Their supports are disjoint, so they lie at Fisher-Rao distance 3.14, the diameter of the simplex. On a task about B they prescribe opposite decisions: the decision optimal under the first belief scores 0.0 in the world of the second, the entire stake of that task. Two beliefs as far apart as the manifold allows are therefore indistinguishable to one task and opposite on another, and the return, which cannot register the difference, cannot serve as the measure of competence.

Two policies with equal return

The agent acts on the world by probing it. A probe is a sensing action with a fixed observation channel, and the model has two: an A -probe that reports A through a confusion channel correct with probability 0.8, and a B -probe that reports B correctly with probability 0.85. Each probe outcome updates the belief by Bayes’ rule, so a plan of probes traces a path across the simplex. Because the world is static and the channels are memoryless, every expectation below is computed exactly by enumerating observation outcomes weighted by their probabilities, without sampling.

Two policies differ only in what they attempt to resolve. The reward-only policy maximizes task-1 return and faces an asymmetry: an A -probe can raise the task-1 value, while a B -probe leaves it un-

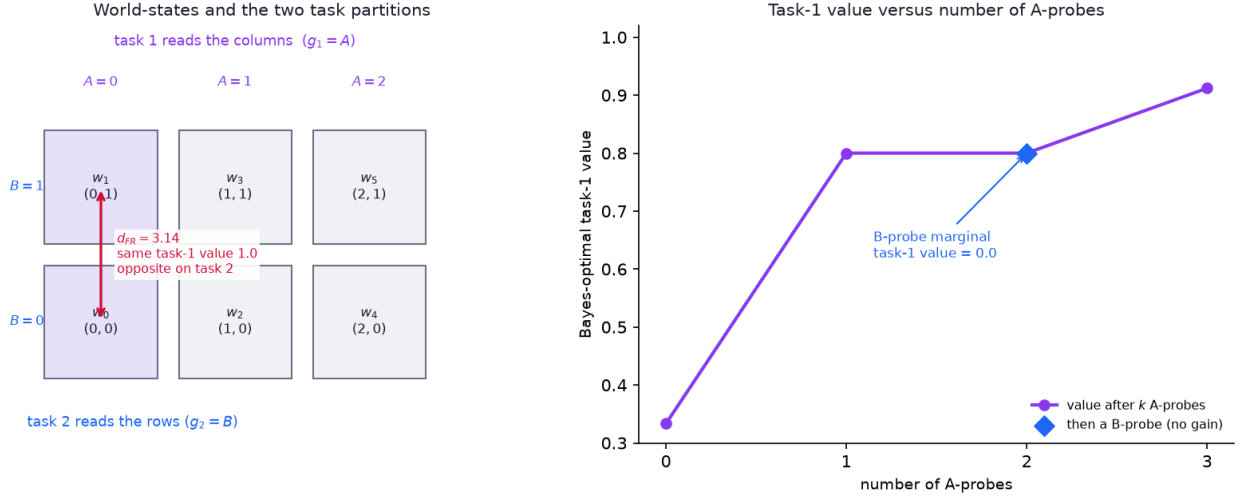


Figure 1: World-states, task partitions, and the value of probing. Left: the 6 world-states as a grid, A across the columns and B down the rows. Task 1 reads only the columns (the partition $g_1 = A$); task 2 reads only the rows ($g_2 = B$). The beliefs at $w_0 = (0, 0)$ and $w_1 = (0, 1)$ lie at Fisher-Rao distance 3.14, the diameter, share the task-1 value 1.0, and take opposite task-2 decisions. Right: Bayes-optimal task-1 value against the number of A -probes, 0.3333 at the prior, 0.8 after one and after two probes, and 0.912 after three; a B -probe added after two A -probes leaves the task-1 value unchanged, a marginal contribution of 0.0, because B lies in the quotient the reward discards.

changed, a marginal task-1 value of 0.0, because B lies in the quotient the reward cannot read. No policy that maximizes task-1 return spends a probe on B . The competent-contact policy chooses probes by expected information gain about the full state w , the classical objective of Bayesian experimental design, equal to the expected reduction in the posterior’s Shannon entropy (Lindley 1956) and to the expected Bayesian surprise, the KL divergence from prior to posterior (Itti and Baldi 2009; Gottlieb et al. 2013). To isolate this difference, both policies receive the same budget of 2 probes on A , and the competent policy additionally spends 2 probes on B .

The return does not distinguish the two policies. B -probes do not affect the A -marginal, so both policies attain the same expected task-1 value of 0.8, a gap of 0.0, and a theory that scores policies by single-task return ranks them as equivalent. With a per-probe cost of 0.05 included in the return, the ranking favours the less competent policy: the reward-only policy nets 0.7 on task 1 and the competent policy 0.6, because the competent policy pays for 2 probes that task 1 does not reward.

The two terminal beliefs differ substantially in geometry. The reward-only policy leaves a terminal belief with Shannon entropy 1.09 nats, most of it the unresolved uncertainty about B ; the competent policy leaves 0.68 nats, with the belief driven further toward a vertex. The Fisher-Rao length from the prior is 1.62 for the competent belief and 1.28 for the reward-only belief. The clearest comparison is representational. For each policy, the terminal belief reached when every probe returns its most likely reading is computed for each true world-state, and the 6-by-6 matrix of Fisher-Rao distances between these beliefs forms a representational dissimilarity matrix in the sense of systems neuroscience (Kriegeskorte et al. 2008; Kriegeskorte and Kievit 2013). Under the competent policy

every pair of world-states remains separated, with minimum separation 2.44. Under the reward-only policy, world-states that share A and differ only in B collapse to separation 0.0, because the policy never probed B ; its representation contains none of the reward-irrelevant distinctions of the world.

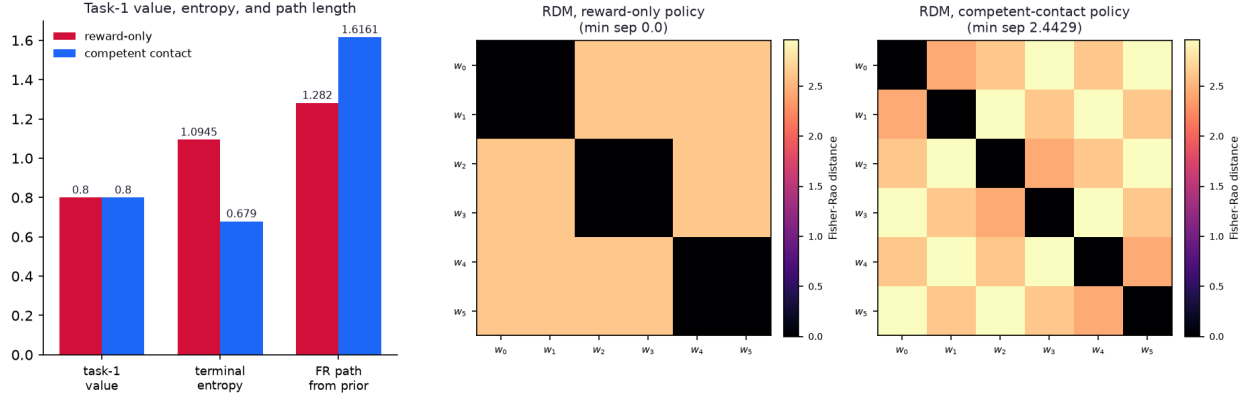


Figure 2: Task-1 value, belief geometry, and representational dissimilarity for the two policies. Left: the reward-only and competent-contact policies attain the same task-1 value 0.8, while the reward-only policy leaves higher terminal entropy (1.0945 against 0.679 nats) and a shorter Fisher-Rao path from the prior (1.282 against 1.6161). Middle and right: representational dissimilarity matrices, the Fisher-Rao distances between terminal beliefs indexed by the true world-state. The reward-only matrix places the three pairs that differ only in B at distance 0.0 (dark diagonal blocks), a minimum separation of 0.0; the competent matrix separates all six states, with minimum 2.4429.

Transfer to a second task

The distinctions discarded by the reward-only policy are structure of the world along a dimension that task 1 does not reward, and a different task rewards that dimension fully. Task 2 depends on the world only through B and arrives after probing is complete, to be answered from the belief the agent already holds, with no further contact. This is the standard transfer situation: an agent that learned about its world for one purpose must act for another without sensing again.

The competent agent transfers and the reward-only agent does not. The reward-only policy never probed B , so its B -marginal remains the uniform prior and its best zero-shot task-2 value is 0.5, chance. The competent policy resolved B to the fidelity of the channel and scores 0.85, a transfer gap of 0.35 equal to the value of the within-class structure the scalar quotient discarded. Averaged over a task family that draws its reward from either attribute with equal probability, the competent agent’s expected value is 0.825 and the reward-only agent’s 0.65. The gap does not reflect general superiority: on task 1 the two are equal, and net of probe cost the competent agent is behind. The gap is entirely the value of the geometry that task 1 did not price and task 2 does.

The competent-contact policy seeks Blackwell-sufficiency for the world instead of for a single task: an experiment sufficient for the full state is preferred in every decision problem the world can pose, which is why it transfers (Blackwell 1953). The information-theoretic drives studied as intrinsic

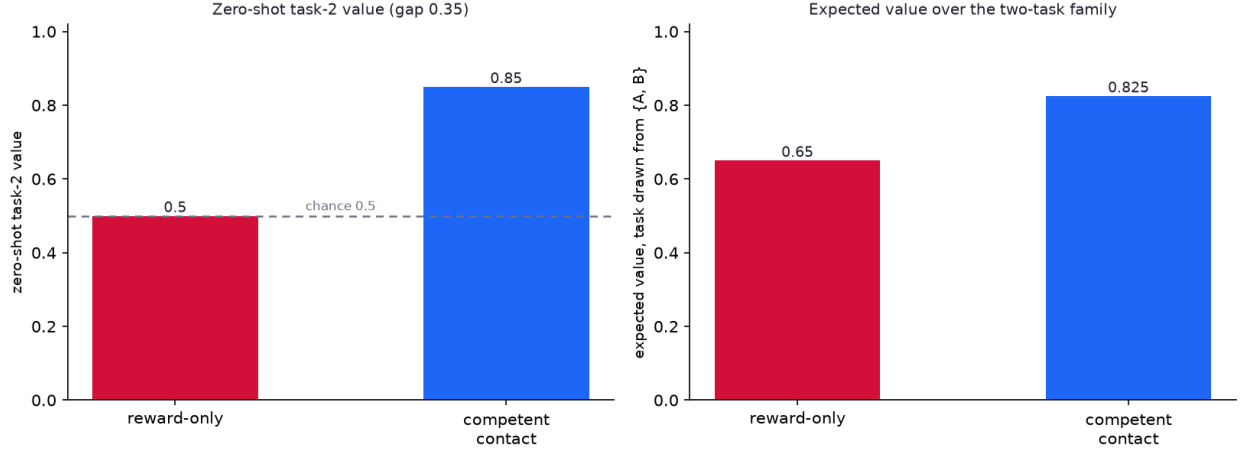


Figure 3: Zero-shot task-2 value and value over a two-task family. Left: task-2 value with no further probing; the reward-only policy scores 0.5, chance, having never resolved B , and the competent policy scores 0.85, a transfer gap of 0.35. Right: expected value over a task family drawing its reward from either attribute with equal probability, 0.65 for the reward-only policy and 0.825 for the competent policy.

motivation are related quantities that differ in specific ways. Empowerment is the channel capacity from an agent’s actions to its later sensor states, the maximum an agent can inject into its own sensory future, and it rewards control (Klyubin et al. 2005; Shannon 1948; Cover and Thomas 2006). Competent contact measures how much the agent’s probing resolves the world’s hidden structure, and an agent can have high empowerment over a world it barely understands. The epistemic term in active inference is closer: expected free energy scores a policy partly by its expected information gain about hidden states, which is the competent-contact objective expressed in variational form (Friston et al. 2015; Tishby and Polani 2011). The present contribution is to identify the object these quantities share, the belief manifold with its Fisher-Rao geometry, and to prove that a scalar return is generically blind to it.

Limitations: the agent’s hypothesis space

The belief manifold is defined over the agent’s own hypothesis space, and the geometry cannot extend beyond it. The world-states w are the possibilities the agent represents, the attributes A and B are the dimensions along which it can distinguish anything, and the Fisher-Rao distances are distances between the agent’s beliefs. An agent whose hypothesis space divided the same substrate along different attributes would occupy a different manifold, and competence measured on one manifold is not commensurable with competence measured on the other. Uexküll placed this at the centre of his biology: every animal lives in an *Umwelt*, a world articulated by its own functional cycles, so that the worlds of the tick, the dog, and the physicist are different worlds built on one physical ground (Uexküll [1934] 2010). Competent contact is contact with the world as the agent represents it, and no agent-independent standpoint exists from which that contact could be scored against the world itself.

This bounds the claim. The geometry does not certify that an agent's hypothesis space is appropriate, and an agent that perfectly resolves its own hypothesis space may be resolving the wrong distinctions. Two stipulations of the model mark the same boundary. First, the reward-relevant and reward-irrelevant partitions were fixed in advance, and the demonstration depends on the difference between them; in a world where every distinction is eventually rewarded there is no discarded geometry, and competent contact and reward maximization coincide. The informative cases are those in which an agent must resolve more than any single task rewards because it does not know which task it will face. Second, the policies were specified by hand to isolate the phenomenon and were not derived as optima; the second A -probe, for example, leaves the task-1 value at 0.8, so the comparison is at a matched probe budget and not between cost-optimal plans. The numbers are properties of the construction and not measurements of any agent. The proposition requires none of these stipulations: wherever reward factors through a coarsening of the world, the return is blind to the finer geometry of belief, and that geometry is what an agent carries from the task it was trained on to a task it was not.

Conclusion

A learner that maximizes a scalar faithful to its training task discards the structure that would support the next task, and it appears most efficient when it is least prepared for a new one. In the model, the reward-only and competent-contact policies are tied at a task-1 return of 0.8, yet only the competent policy retains the within-class distinctions (minimum Fisher-Rao separation 2.44 against 0.0) that raise zero-shot task-2 value from 0.5 to 0.85. The geometry of competent contact on the Fisher-Rao belief manifold records what a scalar objective discards, and the cost of that loss appears whenever the world contains more structure than the current reward pays for.

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